

SQL Set Operations Research – UNION, INTERSECT, and EXCEPT

1. UNION

UNION combines results from two queries and removes duplicate rows.

Syntax

```
SELECT column1, column2  
  
FROM table1  
  
UNION  
  
SELECT column1, column2  
  
FROM table2;
```

Example

```
SELECT Fname AS Name  
  
FROM Employee  
  
UNION  
  
SELECT DependentName AS Name  
  
FROM Dependent;
```

This shows employee names and dependent names without duplicates.

Worksheet Query Builder

```
SELECT Fname AS Name
FROM Employee
UNION
SELECT DependentName AS Name
FROM Dependent;
```

Script Output x Query Result x

SQL | All Rows Fetched: 7 in 0.002 seconds

	NAME
1	Aisha
2	Ali
3	Hamad
4	Maha
5	Omar
6	Salim
7	Sara

2. UNION ALL

UNION ALL combines results but keeps duplicate rows.

Example

```
SELECT Fname AS Name
```

```
FROM Employee
```

```
UNION ALL
```

```
SELECT DependentName AS Name
```

```
FROM Dependent;
```

Difference:

UNION removes duplicates.

UNION ALL keeps duplicates.

The screenshot shows a database query builder interface. The top section is titled 'Worksheet' and 'Query Builder'. The query text is as follows:

```
SELECT Fname AS Name
FROM Employee
UNION ALL
SELECT DependentName AS Name
FROM Dependent;
```

The bottom section shows the 'Query Result' tab. It displays a table with 7 rows and 1 column named 'NAME'. The data is as follows:

	NAME
1	Ali
2	Sara
3	Omar
4	Maha
5	Aisha
6	Hamad
7	Salim

3. INTERSECT

INTERSECT returns only rows that appear in both queries.

Syntax

SELECT column1

FROM table1

INTERSECT

SELECT column1

FROM table2;

Example

SELECT Dnum

FROM Employee

INTERSECT

SELECT Dnum

FROM Project;

This shows departments that have both employees and projects.

The screenshot shows a SQL Query Builder window with two tabs: 'Worksheet' and 'Query Builder'. The 'Query Builder' tab is active, displaying the following SQL query:

```
SELECT Dnum
FROM Employee
INTERSECT
SELECT Dnum
FROM Project;
```

Below the query editor, there is a 'Script Output' tab and a 'Query Result' tab. The 'Query Result' tab is active, showing the results of the query. The status bar indicates 'All Rows Fetched: 3 in 0.004 sec'. The results are displayed in a table with two columns: 'DNUM' and an unnamed column.

	DNUM	
1	10	
2	20	
3	30	

4. EXCEPT

EXCEPT returns rows from the first query that do not appear in the second query.

In Oracle SQL, use **MINUS** instead of **EXCEPT**.

Syntax

SELECT column1

FROM table1

MINUS

SELECT column1

FROM table2;

Example

SELECT Dnum

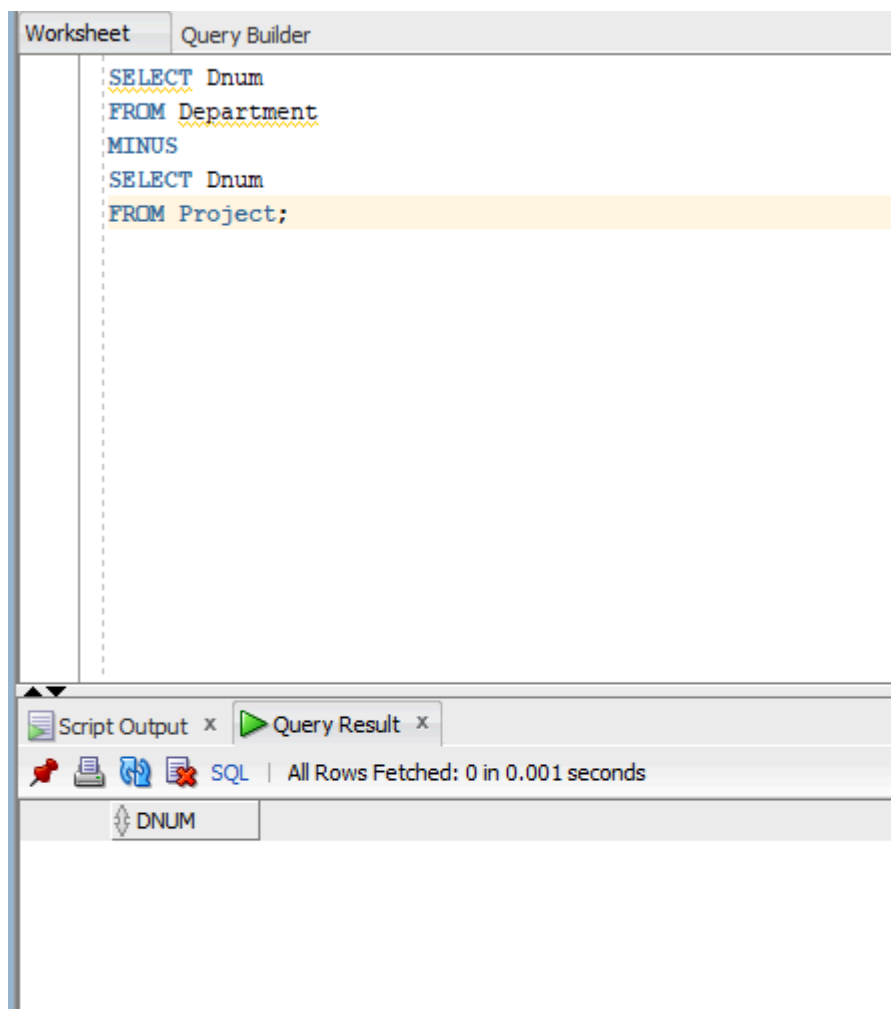
FROM Department

MINUS

SELECT Dnum

FROM Project;

This shows departments that have no projects.



Worksheet Query Builder

```
INSERT INTO Department VALUES (40, 'Marketing', NULL, NULL);  
COMMIT;
```

For test

Worksheet Query Builder

```
SELECT Dnum  
FROM Department  
MINUS  
SELECT Dnum  
FROM Project;
```

Script Output x Query Result x

 SQL | All Rows Fetched: 1 in 0.002 sec

	DNUM
1	40